

Efficient Frontier

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1 Calculation of the Mean - Variance equation using Lagrange multipliers

1.1 Equation and constraints

$$\min_w w^\top \Sigma w$$

Subject to:

$$w^\top \mu = r^*, 1^\top w = 1$$

1.2 Construction of the Lagrangian

$$\mathcal{L}(w, \lambda_1, \lambda_2) = w^\top \Sigma w - \lambda_1(w^\top \mu - r^*) - \lambda_2(1^\top w - 1)$$

1.3 Matrix derivatives and equal to 0

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial w} &= 2\Sigma w - \lambda_1 \mu - \lambda_2 1 = 0 \\ \frac{\partial \mathcal{L}}{\partial \lambda_1} &= w^\top \mu - r^* = 0 \\ \frac{\partial \mathcal{L}}{\partial \lambda_2} &= 1^\top w - 1 = 0\end{aligned}$$

1.4 Isolate the unknown vector

$$\begin{aligned} 2\Sigma w - \lambda_1 \mu - \lambda_2 1 &= 0 \\ 2\Sigma w &= \lambda_1 \mu + \lambda_2 1 \\ \Sigma w &= \frac{\lambda_1}{2} \mu + \frac{\lambda_2}{2} 1 \end{aligned}$$

We can redistribute the right side of the equation into two matrices.

$$A = (\mu \ 1), \gamma = \begin{pmatrix} \lambda_1/2 \\ \lambda_2/2 \end{pmatrix}$$

Continue isolating

$$\begin{aligned} \Sigma w &= A\gamma \\ w &= \Sigma^{-1}A\gamma \end{aligned}$$

1.5 Combining both constraints

Both constraints have the form $a^\top b$, and since a and b are vectors of the same size, $a^\top b = b^\top a$. With this in mind, we can rearrange the equations so that the weight vector appears on the same side.

In this case, we will rearrange them so that it appears on the right-hand side; the constraint $1^\top w = 1$ already follows this format, so we only need to rearrange the constraint $w^\top \mu = r^*$.

$$w^\top \mu \Rightarrow \mu^\top w = r^*$$

This process allows us to rearrange $1^\top$ and $\mu^\top$ in a matrix, taking advantage of the fact that both are $1 \cdot n$ vectors.

$$A^\top = \begin{pmatrix} 1^\top \\ \mu^\top \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ \mu_1 & \mu_2 & \mu_3 \end{pmatrix}$$

At the same time, our objectives 1 and r^* can be accommodated in another vector.

$$b = \begin{pmatrix} 1 \\ r^* \end{pmatrix}$$

Finally, the equation with both restrictions becomes

$$A^\top w = b$$

1.6 Replace w into our constraint and solve for γ

$$\begin{aligned} A^T w &= b \\ A^T (\Sigma^{-1} A \gamma) &= b \\ (A^T \Sigma^{-1} A) \gamma &= b \\ \gamma &= (A^T \Sigma^{-1} A)^{-1} b \end{aligned}$$

1.7 Replace γ in w

$$w = \Sigma^{-1} A (A^T \Sigma^{-1} A)^{-1} b$$

1.8 Visualization

Analytical frontier allows short selling. Monte Carlo cloud is long-only.

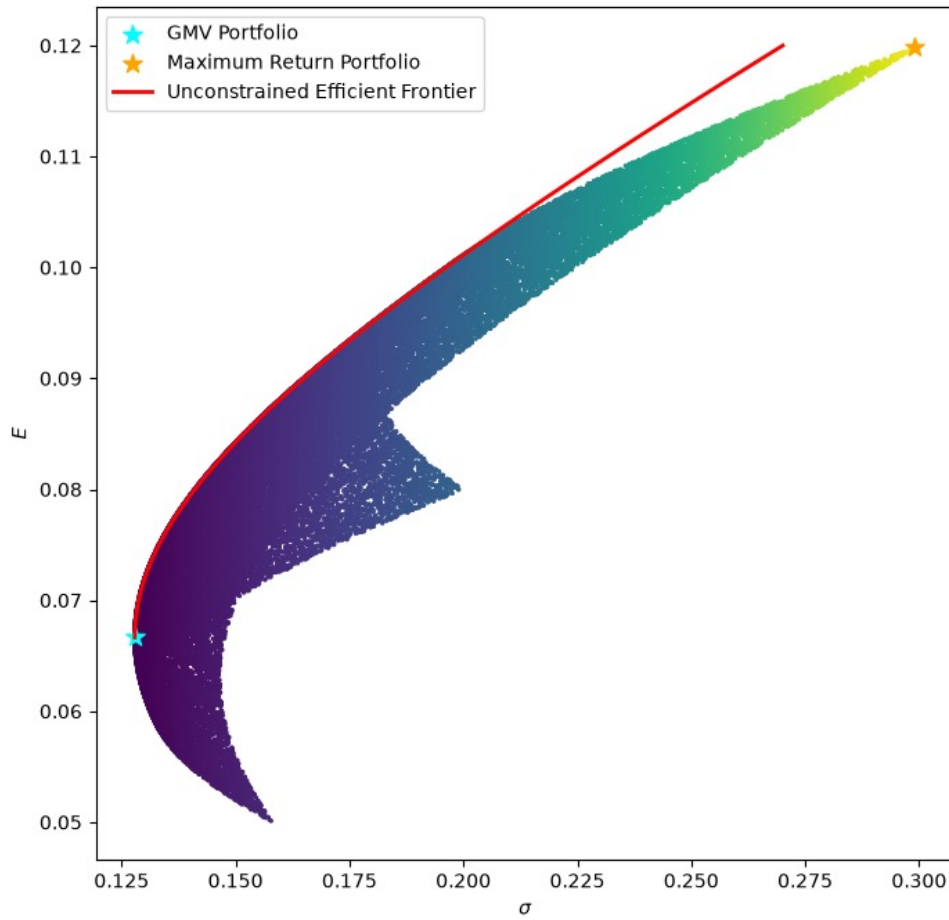


Figure 1: 50,000 Long-Only Portfolios and Unconstrained Efficient Frontier